

DIGITAL LOGIC DESIGN

(EE 223)
DE 44 MECHATRONICS
SYNDICATE - C



Line Following Robot Using Digital Logic

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Abstract and Objectives:

Developing a line-following robot for the "Ready to Race" division of the National Engineering Robotics Contest (NERC) 2024 is the goal of this Digital Logic Design project. The main objective is to make the robot capable of navigating different intersections and obstacles. This project is to be completed using only self-built logic circuits implemented on PCBs/Vero boards, with no dependence on microcontrollers. Using what we learned in class and supplementing it with our own research from sites like YouTube and Google, we used Proteus to construct and simulate a circuit, evaluate its performance to get the desired result, and resolve any problems that arose. The final goal is for the robot to navigate through intersections with ease, making the appropriate turns according to the map that is supplied.

List of components used:

1. AND Gate – IC 7408
2. OR Gate – IC 7432
3. XOR Gate – IC 7486
4. 8 to 1 Multiplexer - IC 74151
5. 2 to 1 Multiplexer – IC 74157
6. Timer – IC NE555
7. Multivibrator - IC 74123
8. 4 bit asynchronous ripple counter – IC 7493
9. Resistors
10. Potentiometers
11. IR Sensors
12. N20 Motors

Initial Logic:

Our primary objective was to create a logic system enabling a robot to autonomously follow a designated line path. After conducting thorough research, we chose to implement the line-following logic using a single integrated circuit (IC) known as L293D. The core principle of this logic was reliant on the functionality of sensors placed on either side of the robot. If the right sensor

detected the absence of the line, the right motor would be deactivated. Similarly, when the left sensor registered the absence of the line, the left motor would cease functioning. This configuration allowed the robot to effectively track and follow the line's course, resulting in a functional line-following robot.

Arena problem faced:

Upon analyzing the arena, we observed that it did not consist of a straightforward line but instead encompassed numerous junctions. This realization prompted us to reassess our initial logic and devise a new approach capable of detecting and navigating these junctions. Our focus shifted towards creating a logic system that could not only identify but also count the junctions, enabling the robot to make specific turns at these critical points. Our research led us to explore the application of a counter mechanism to tally the junctions encountered by the robot.

Ripple Counter?

Initially, the CD4017 counter seemed promising; however, we discovered limitations in its ability to count up to the desired number of junctions required for our purposes. Consequently, we sought a more suitable solution and identified the need for a 4-bit counter. Subsequently, we opted for the asynchronous ripple counter, specifically the 7493, as it met our criteria for effectively counting and addressing the junctions encountered by the robot during its navigation within the arena.

Counter problem

The primary challenge encountered with the counter revolved around the necessity for a pristine clock pulse—void of any noise or unwanted fluctuations—to prevent false counts. Initially, while employing toggle switches on the trainer board to generate the clock pulse for the counter, the switching action occasionally produced multiple falling edges, leading to inaccurate counts. The detection of these false counts perplexed us initially, taking nearly a week to discern the underlying cause. Transitioning to utilizing push buttons on the

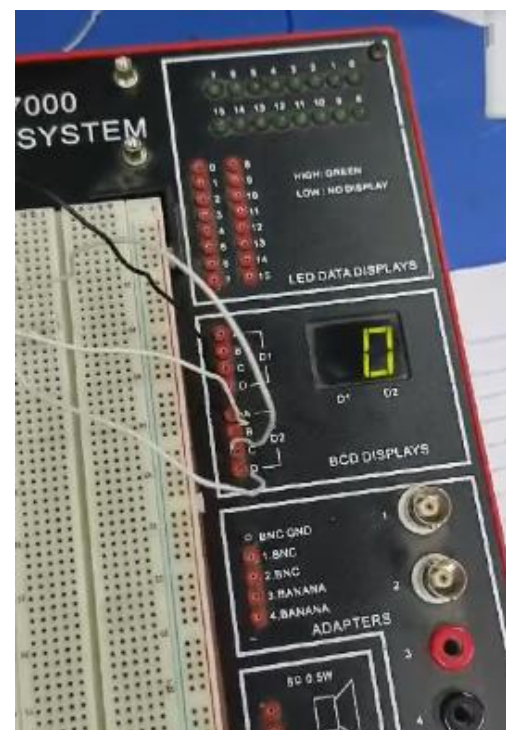


Figure 1.1 Attempt with toggle switches

trainer board rectified this issue by providing a cleaner pulse, ensuring accurate counts. However, replicating this manual process in the circuit was impractical, necessitating an alternative solution to guarantee a clean pulse for the counter. To achieve this, we incorporated a timer into the circuit with the shortest delay setting, allowing a brief pause between sensor readings. This implementation ensured a clean and distinct interval between successive sensor inputs, effectively mitigating the possibility of introducing garbage values and maintaining the accuracy of the counter's readings.



Figure 1.2 Attempt with Push Buttons

Multivibrator

In our efforts to create a multivibrator system for controlling the motor's timer, initially utilizing the NE555 IC, we encountered a persistent issue during the Proteus implementation—while the motor would stop as intended, it failed to restart. Research revealed that the NE555 monostable timer was not providing a short pulse, resulting in a constant pulse that inhibited the timer from restarting the motor. Recognizing this inefficiency, we sought an alternative and found the 74123 IC, known for generating short pulses, thus facilitating the successful activation and deactivation of the timer. Figure 1.3 illustrates the truth table of the 74123 IC, demonstrating its capability to produce a single short pulse. Implementing capacitors and resistors in conjunction with the 74123 IC allowed us to establish the required delay within the timer, ensuring precise control over the motor's activation and deactivation periods.

INPUTS			OUTPUTS	
CLEAR	A	B	Q	$\bar{Q}$
L	X	X	L	H
X	H	X	L [†]	H [†]
X	X	L	L [†]	H [†]
H	L	↑	⌋	⌋
H	↓	H	⌋	⌋
↑	L	H	⌋	⌋

Figure 1.3 Truth Table of 74123

Multiplexer

To strategically navigate the maze while bypassing certain junctions and precisely turning at specific points, our focus shifted towards integrating multiplexers (mux). Understanding their function as data selectors, we devised a logical approach: utilizing two 8-into-1 muxes for both left and right turns. By meticulously studying the counter's data sheets, we identified the specific counts corresponding to the junctions where turns were required, notably the 4th, 6th, 10th, and 14th counts. For the left turns at the 4th and 6th counts, we directed the three relevant bits from the counter to the select lines of the left mux. Similarly, for the right turns at the 10th and 14th counts, we directed the corresponding bits from the counter to the select lines of the right mux. This meticulous integration and utilization of multiplexers allowed for precise decision-making at critical junctions, enabling the robot to navigate the designated route effectively while executing accurate turns as needed. Adapting to the physical layout of the arena presented a challenge as the initially anticipated junction layout differed from the physical reality. The provided arena layout lacked a specific junction line crucial for our initial logic, prompting us to rethink our approach. Originally intending to turn at the 4th, 5th, 9th, and 13th junctions based on the provided layout, the realization of additional junctions upon physical implementation forced us to recalibrate our logic for turns. Consequently, we adjusted the logic to execute turns at the 4th, 6th, 10th, and 14th junctions upon physically observing the expanded junction configuration within the arena. This necessitated a shift in our robot's decision-making logic, prompting the incorporation of new junction points for accurate navigation within the arena's revised layout. Integrating multiplexers then became essential to facilitate precise turns at these updated critical junctions, ensuring the robot followed the modified path effectively.

Turning problem:

The subsequent challenge emerged when the output from the 8-to-1 multiplexer directly controlled the motors. Upon reaching the 4th count, the motor appropriately switched off, but reactivating it posed an obstacle. Although the motor should have turned on upon reaching the 5th junction, it perpetually failed to do so, causing the robot to circle endlessly without progressing to the next junction. In response, a solution was envisioned involving the utilization of a timer

mechanism. This proposed approach aimed to temporarily deactivate the motor and trigger its reactivation after a predetermined interval of time elapsed, effectively resolving the issue of the motor's failure to restart at the designated junctions.

Timer:

In our pursuit to resolve the motor reactivation issue, research led us to identify the suitability of employing a monostable timer for the task at hand. Implementing two timers became imperative for distinct purposes: one for regulating the duration of the turning period, allowing the motor to stop and execute the turn for a specific interval, while the other aimed to prevent the inadvertent counting of spurious sensor values during this turning phase. The latter timer was designed to inhibit the counter from incrementing while the robot executed turns. However, an obstacle surfaced when the turning timer activated on the 4th count but failed to deactivate afterward. This issue stemmed from the timer's requirement for a trigger—a short, singular pulse—while we were supplying it with a continuous gate pulse. Initially, our attempted resolution involved the use of a 4060 IC as a frequency divider, yet its consistent fluctuations exacerbated the issue. Subsequently, we turned to the 74123 multivibrator IC, recognized for its capability to produce a singular, momentary pulse—either a rising or falling edge pulse. Leveraging this IC, we successfully addressed the problem by providing the monostable timer with the required short pulse, enabling effective control over the motor's activation and deactivation at the specified junctions during the robot's navigation.

Final touches to logic

After conducting a simulation of the finalized circuit, we discovered that the second timer, intended to address spurious sensor readings, proved to be unnecessary as the circuit functioned effectively without its inclusion. Consequently, we streamlined our design, simplifying the circuit configuration. Introducing a potentiometer into the timer setup enabled us to adjust and fine-tune the timer's duration, offering increased flexibility by allowing us to modify the timer values as needed. With the circuit functionality largely optimized, the remaining task was integrating a motor driver into our setup. This step proved to be time-consuming, yet we successfully accomplished it by implementing the L293D

motor driver, providing the necessary control and power regulation for the motors, thereby completing the integration of all essential components into our circuit for the line-following robot.

Chassis

Once we had finalized and thoroughly tested our logic using Proteus simulation (as depicted in Figure 1.4), we proceeded to construct the hardware. Initially, we opted for an acrylic sheet measuring 3mm in thickness (illustrated in Figure 1.5). After cutting it to the required dimensions, we mounted the motors and a castor wheel onto the sheet to serve as the chassis. Unfortunately, the 3mm acrylic sheet proved insufficiently sturdy, leading to breakage after a few days of use. As a result, we switched to a more robust material, utilizing a 5mm thick alternative (as shown in Figure 1.6) to create a more durable chassis for the line-following robot.



Figure 1.5 Initial chassis design

Sensor Calibration

To calibrate the sensors for optimal performance, we initially positioned them over the white line and observed whether they returned a value of zero (shown in figure 1.7). If the sensors did not register zero while placed on the white surface, we adjusted their sensitivity settings to ensure they could accurately detect the white line. Similarly, we repeated this calibration process for the black color by placing the sensors on the black surface and adjusting their sensitivity until they reliably detected the black color. This meticulous calibration procedure aimed to fine-tune the sensors' sensitivity, allowing them to distinguish between the contrasting colors of the line accurately, thereby enhancing the precision of the robot's line-following capability.

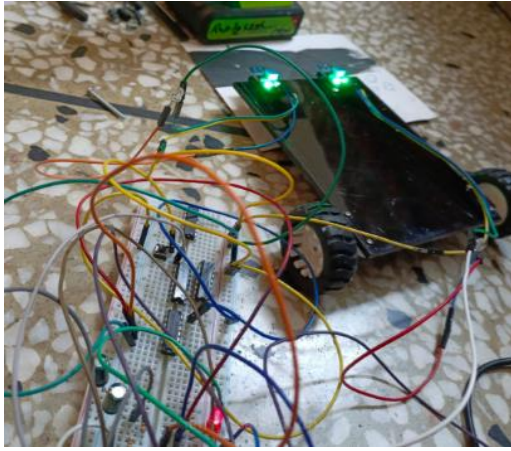


Figure 1.7 Sensor Calibration



Figure 1.6 Final chassis design

Logic Implementation

Testing the logic step by step on the breadboard, we meticulously ensured each component's functionality on the trainer board to validate our progress. Beginning with the sensors, we utilized LED displays as an output to verify their accuracy in providing the desired output (shown in figure 1.8). Subsequently, integrating the counter and its timer allowed us to assess the correctness of the counting mechanism. During testing, we encountered occasional issues with improper counting, which we traced back to wire movements causing disruptions.

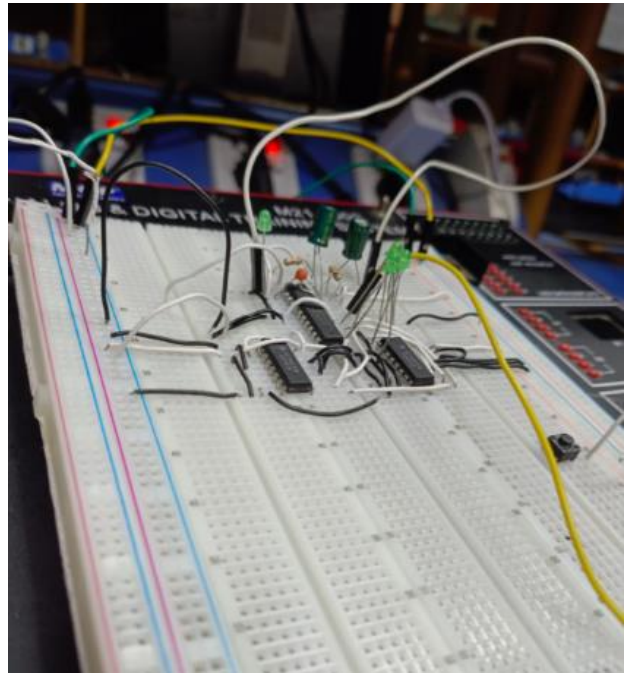


Figure 1.8 Breadboard implementation

Ensuring stable wiring resolved these inconsistencies, ensuring accurate counting within the system. Progressing to the implementation of the multiplexers directing the left and right motors and the subsequent demultiplexer for motor control, we systematically checked and validated their functionality. As each component successfully fell into place, we approached the final stage: integrating the motor driver, marking the completion of our meticulous step-by-step implementation of the entire logic onto the breadboard.

Motor Driver

During the motor driver implementation phase, meticulous steps were taken to ensure proper functionality and address unexpected issues. Initial tests with self-assembled batteries operating at 12V resulted in a burnt left motor, signaling the need for voltage regulation. Subsequent investigation revealed that the high voltage had caused motor damage. Employing a voltage regulator to decrease the supply voltage to a safer 5V range resolved the issue of excessive speed but led to the discovery that the right motor failed to operate. Detailed examination with a digital multimeter (DMM) exposed a voltage drop, indicating a potential issue with current supply. Further research confirmed that the motor required higher amperage, whereas the 2-to-1 mux was delivering milliampere-level outputs. To resolve this, integration of the L293D IC served as an intermediary between the 2-to-1 mux and the motor driver (L298N), rectifying the functionality for both motors. However, the motors were running at excessively high speeds. Implementing pulse-width modulation (PWM) using a low resistance and low-value potentiometer allowed effective control over motor speeds. With these

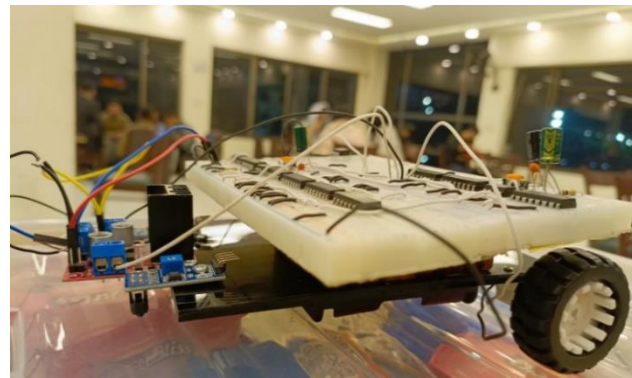


Figure 1.9 Final Robot

modifications in place, the motors operated smoothly with adjustable speeds, setting the stage for comprehensive testing to ensure seamless coordination among components for the intended functionality of the line-following robot. The final robot is shown in figure 1.9.

Testing

During the testing phase, recalibration of the sensors became necessary due to highly reflective surfaces affecting their accuracy. However, a significant challenge arose from the arena's underlying terrain, which was strewn with rocks, leading to deviations from the intended path. While the robot successfully navigated the maze, the irregular terrain posed a persistent issue: the robot's tires frequently became lodged among the rocks, causing delays in executing turns and

subsequently deviating from the designated path. Regrettably, this obstacle couldn't be effectively mitigated, resulting in the robot's completion rate of the maze limited to around 3 out of 10 attempts. Despite the robot's ability to maneuver through the maze, the unpredictable nature of the rocky terrain continually disrupted its movements, leading to incomplete or erratic turns, ultimately impacting the robot's overall performance within the arena.

Conclusion

In a nutshell, the culmination of this project has been a testament to the intricate nature of robotics and the challenges inherent in real-world implementations. As we navigated through the complexities of designing a line-following robot, we were not only tasked with understanding and implementing intricate circuitry but also encountered unforeseen hurdles, such as sensor calibration issues and the unpredictable terrain within the arena. These challenges demanded constant problem-solving skills and adaptability. The process taught us the invaluable lesson of meticulously managing time and resources, ensuring each component was rigorously tested and integrated. Despite facing setbacks with the rocky terrain disrupting the robot's performance, this project underscored the importance of resilience and perseverance in the face of unforeseen obstacles. Overall, the project was a rich learning experience that honed our problem-solving abilities, emphasized the significance of meticulous planning, and highlighted the need for adaptability when navigating real-world complexities in robotics projects.

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